

CIA 1

L&T: Node and Express.js

Theory and Practical

Theory: Set A

Practical: Set 1

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CIA 1 Theory

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Node & Express JS. CIA 1 (Theory) SET A

Q1 JavaScript code can be run directly inside a browser without needing a separate compilation step before execution.

Q2 Write a program to declare a variable 'price' with value 250, then increase it by 50 and print the new value.

⇒ Program:-

```
let price = 250;  
price = price + 50;  
console.log("New value of price:" + price);
```

Q3 The value 'undefined' in JavaScript indicates that a variable has been declared but not assigned with a value.

Q4 Write a program to check if two numbers are equal using both == and ===, and print both results.

⇒ Program:-

```
let a = 10;  
let b = 10.0;  
  
if (a == b)  
  console.log("True - using ==");  
else  
  console.log("False - using ==");  
  
if (a === b)  
  console.log("True - using ===, checks data type also");  
else  
  console.log("False - using ===, checks data type");
```

Result:-

True - using ==

False - using ===, checks data type

- Q5. A function that is defined without a name and assigned to a variable is called a anonymous function.

CIA 1 Practical

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Node & Express JS - CIA 1 Practical - Set 1

→ Task 1 : Student Record System - Welcome Message

```
console.log("Welcome to Eduhog: Student Record System.");
```

Output:- node task1.js.

Welcome to Eduhog: Student Record System

→ Task 2: Student Record System - Declare and update variables.

```
const SysName = "Eduhog: Student Record System";
```

```
let totalStudents = 2560;
```

```
console.log(`Welcome to ${SysName}`);
```

```
console.log(`Total Students: ${totalStudents}`);
```

Output:-

Welcome to Eduhog: Student Record System

Total Students: 2560

→ Task 3: Identify Data types

```
let StudentName = "Christy Joe";
```

```
let StudentID = 1242;
```

```
let isEnrolled = true;
```

```
console.log("Value:" + StudentName + " | Type:" + typeof StudentName);
```

```
console.log("Value:" + StudentID + " | Type:" + typeof StudentID);
```

```
console.log("Value:" + isEnrolled + " | Type:" + typeof isEnrolled);
```

Output:-

Value: Christy Joe | Type: string

Value: 1242 | Type: number

Value: true | Type: boolean

→ Task 4: Perform a Calculation

```
let m1 = 45;
```

```
let m2 = 78;
```

```
let m3 = 91;
```

```
let average = (m1 + m2 + m3) / 3;
```

```
console.log(`Average marks of three students is: ${average}`);
```

Output:-

Average marks of three students is: 71.333333

→ ~~Task~~ Task 5: Classify a Record

```
let marks = 50;
console.log("Student Marks: " + marks);
if (marks >= 40)
  console.log("Pass");
else
  console.log("Fail");
```

Output:-

Student Marks: 50

Pass.

→ Task 6: Iterate over Records

```
let students = [
  { id: 101, name: "Samuel Johnson", percentage: 85, grade: 'A' },
  { id: 102, name: "Lily", percentage: 75, grade: 'B' },
  { id: 103, name: "Dino", percentage: 68, grade: 'C' },
  { id: 104, name: "Meera", percentage: 50, grade: 'D' }
];

for (student of students) {
  console.log("=====");
  console.log("Student ID: " + student.id);
  console.log("Student Name: " + student.name);
  console.log("Student Percentage: " + student.percentage);
  console.log("Grade: " + student.grade);
}
```

Output:-

```
=====
Student ID: 101
Student Name: Samuel Johnson
Student Percentage: 85
Grade: A
=====
Student ID: 102
Student Name: Lily
Student Percentage: 75
Grade: B
=====
Student ID: 103
Student Name: Dino
Student Percentage: 68
Grade: C
=====
Student ID: 104
Student Name: Meera
Student Percentage: 50
Grade: D
```

→ Task 7: Skip or Stop Early

```
for (student of students) {  
  if (student.marks >= 85) marks  
    continue;  
  else if (student.marks < 60)   
    break;  
  else  
    console.log("-----");  
    console.log(`ID: ${student.id}`);  
    console.log(`Name: ${student.name}`);  
    console.log(`Marks: ${student.marks}`);  
    console.log(`Grade: ${student.grade}`);  
}
```

Output:-

```
-----  
ID: 102  
Name: Lily  
Marks: 75  
Grade: B  
-----
```

```
ID: 103  
Name: Dino  
Marks: 68  
Grade: C
```

→ Task 8: Add & Find

```
students.push({id: 105, name: "David", marks: 95, grade: 'A'});  
let maxMarks = 0; for (student of students) {  
  if (student.marks > maxMarks)  
    maxMarks = student.marks;  
}  
console.log(`The Highest Marks: ${maxMarks}`);
```

Output:-

The Highest Marks: 95.

→ Task 9: Remove & Sort

```
student.pop(); // Removes the last element  
students.sort((a, b) => b.marks - a.marks);  
console.log("Highest to lowest in terms of marks:");  
for (student of students) {  
  console.log(`Student ID: ${student.id}, Name: ${student.name},  
    Marks: ${student.marks}`);  
}
```

Output:-

Highest to lowest in terms of marks

Student ID: 101, Name: Samuel Johnson, Marks: 85

Student ID: 102, Name: Dino, Marks: 82

Student ID: 103, Name: Lily, Marks: 75

Student ID: 104, Name: Meera, Marks: 50

→ Task 10: Create and Print an Object

```
const student = {  
  id: '102',  
  name: "Kate",  
  age: 20,  
  major: "Criminal Psychology",  
  marks: 75  
};
```

```
console.log("Student Details");  
console.log('ID: ${student.id}');  
console.log('Name: ${student.name}');  
console.log('Age: ${student.age}');  
console.log('Major: ${student.major}');  
console.log('Marks: ${student.marks}');
```

Output:-

Student Details

ID: 102

name: Kate

age: 20

major: Criminal Psychology

marks: 75.